

RESEARCH ARTICLE

Self-Releasing Vanillin: An Eco-Friendly Self-Polishing Coating Designed for Marine Antifouling

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ABSTRACT

Biological fouling will cause severe damage to the functionality of marine underwater facilities, shorten their service life, impede their normal operation, and trigger a biological invasion event. The traditional marine antifouling coatings release heavy metal elements, which cause harm to the marine environment. To overcome this issue, a self-polishing antifouling resin based on vanillin methacrylate (VMA) was prepared via the free radical polymerization method. In seawater, the VMA-based coatings achieve self-polishing through the hydrolysis of the phenolic ester group. The hydrolysis mechanism of VMA was calculated using density functional theory by Cp2k software, and the results show that VMA achieves one-step hydrolysis by the S_N2 mechanism. It exhibited antibacterial and environmentally friendly antifouling properties by releasing vanillin, which is a natural antifouling active substance. The antibacterial, diatom adhesion inhibition, and the 9-month marine field test demonstrated the excellent antifouling performance of VMA-based coatings. It provides a novel concept for environmentally friendly self-polishing coatings and has a promising application prospect.

1 | Introduction

Adhering to various surfaces underwater is a pivotal survival strategy for bacteria, diatoms, mussels, and many other marine organisms [1], which caused marine biofouling. Marine biofouling is a substantial global problem facing mankind. It affects the development of the shipping industry, promotes greenhouse gas emissions, and brings unknown risks of biological invasion [2, 3]. Marine antifouling (AF) coatings can effectively overcome this problem [4], such as low surface energy coatings [5], slippery liquid-infused porous surfaces [6, 7], zwitterionic surfaces [8, 9], self-polishing coatings

(SPCs) [10, 11], etc. In these kinds of AF coatings, the SPCs are the most prevalently utilized due to their efficacy, facile application process, and economy [12]. However, since highly toxic tributyltin-based SPCs were banned [13], silicone-based [14], copper-based [10], or zinc-based [15] acrylate polymers become a more common alternative. These SPCs usually have hydrolyzable side groups [12], which means that it can be hydrolyzed in seawater, rendering the coating surface more hydrophilic and promoting its shedding, finally constituting a self-renewing and unstable surface to prevent the adhesion of fouling organisms [16]. Nevertheless, most acrylate polymers rely on metal fillers such as Cu_2O as AF agents [17–19] and

exhibit poor marine AF performance under static conditions [20, 21]. Therefore, it is imperative to develop environmentally friendly acrylate polymers with enhanced AF efficacy in static environments. The introduction of environmentally friendly substances and the preparation of self-polishing monomers could be one approach.

Vanillin is an economical and environmentally friendly biobased material, comes from a wide range of sources, and has a pleasant aroma [22–25]. It is widely used in the food industry [26], cosmetics industry [27], and pharmaceutical industry [28], etc. More importantly, vanillin's antibacterial properties make it a potential AF active substance. Chen et al. [29] studied the antibacterial mechanism of vanillin against *Escherichia coli* (*E. coli*). The results show that vanillin kills *E. coli* by damaging cell membranes and disrupting energy metabolism. Elena et al. [30] selected a variety of bacteria and compared the antibacterial properties of capsaicin, eugenol, and vanillin. Results prove that vanillin's antibacterial property is similar to capsaicin and eugenol. The introduction of capsaicin [31] and eugenol [32–34] into SPCs has been reported, and it has been demonstrated that this approach provides a new anti-fouling strategy and shows good results in real-world fouling. Therefore, we propose a new type of vanillin-based acrylate polymers and try to solve these contradictions. To our knowledge, this is the first time vanillin has been introduced into SPCs.

In this work, the vanillin methacrylate (VMA) monomer was successfully prepared and a series of VMA random copolymers (VMPs) with different VMA content were prepared by free radical polymerization. It is a self-polishing polymer based on phenol ester hydrolysis, which proved to be feasible [32–34]. As far as we know, there are some researches on the hydrolysis mechanism of ester [35, 36] but there are few studies on self-polishing polymers [37] specifically studied for phenolic ester hydrolysis. Therefore, VMA as a kind of typical phenolic ester-based hydrolyzed resin was used to propose a possible hydrolysis process of phenol esters using Cp2k [38, 39]. The possible molecular structures were optimized in the hydrolysis of VMA and compared them with triisopropylsilyl methacrylate (TISMA) to analyze the differences in the hydrolysis mechanisms. Antibacterial tests, diatom adhesion inhibition tests, and marine field tests prove that VMA-based SPCs expected to become a novel and environmental friendly AF coating and this strategy has a good application prospect.

2 | Experimental Section

2.1 | Materials

Vanillin was purchased from Heowns Biochem Technologies LLC, Tianjin, methacryloyl chloride and benzoyl peroxide (BPO) were purchased from Shanghai Macklin Biochemical Technology Co. Ltd., hexafluorobutyl methacrylate (HFBMA) was purchased from Harbin Xeogia Fluorin&Silicon Chemistry Co. Ltd., 2-methoxyethyl acrylate (MEA) was purchased from Aladdin Company. 2,2'-Azobis(2-methylpropionitrile) (AIBN), methyl methacrylate (MMA), and butyl methacrylate (BA) were purchased from Tianjin Damao Chemical Reagent Factory, triethylamine, N,N-dimethylformamide

(DMF), dichloromethane, ethanol, and ethyl acetate were purchased from Tianjin Fuyu Fine Chemical Co. Ltd. The raw materials and reagents are analytical reagent and have not been further purified.

2.2 | Synthesis of VMA

Five grams of vanillin and 7.8g triethylamine were dissolved in 50g dichloromethane in advance. The temperature of the system was kept between 0°C and 5°C by ice salt bath, and the mixture was continuously stirred. 6.3g methacryloyl chloride was dissolved in 20g dichloromethane and slowly added to the system through a constant pressure drip funnel for 2h. After the drip was completed, kept stirring for 12h. Upon completion of the reaction, the triethylamine hydrochloride precipitated by filtration, the solvent was removed by rotary evaporation. The oil product obtained was extracted by ethyl acetate–water system. The ethyl acetate of organic layer was removed by spin evaporation, then recrystallized in ethanol–water system, and VMA was obtained after drying.

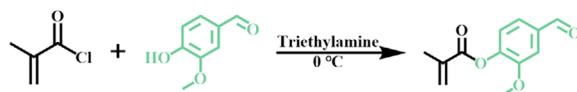
2.3 | Synthesis of Vanillin Methacrylate Copolymers

Vanillin methacrylate copolymers (VMPs) were synthesized by free radical polymerization. First, DMF (90g) was added to a four-necked flask with a mechanical agitator, a condenser, a thermometer, and a constant-pressure drip funnel. Heat DMF in oil bath to 90°C. After that, a certain amount of MMA, BA, HFBMA, MEA, VMA, and initiator AIBN (1.00g) were weighed and fully mixed, then added to the constant pressure-dropping funnel. In the nitrogen atmosphere, it was slowly added to the four-necked flask for 3 h, and then BPO (0.75 g) was dissolved in DMF (10g) and added to the flask. Finally, VMPs resin was obtained by continuous stirring reaction for 3 h to fully polymerize. The synthesis route of VMA and VMPs is shown in Scheme 1. The dosage of each monomer is shown in Table 1.

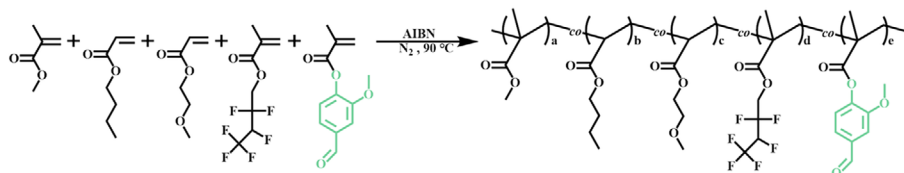
2.4 | Characterization

The Fourier-transform infrared (FTIR) spectra measurements were recorded on an FTIR spectrometer (Spectrum 100, PerkinElmer, USA) in the 400–4000 cm^{−1} by using the KBr-disk method. The proton nuclear magnetic resonance (¹H NMR) spectra were acquired on a BRUKER AVANCE 400 NMR spectrometer (400/500MHz) using CDCl₃ as solvent. The scanning electron microscopy (SEM) images were obtained from a field emission scanning electron microscope (JSM-6480A, JEOL, Japan). The hydrolysis properties are tested through coating the polymer on a glass slide. The coatings are immersed in artificial seawater and dried for testing after a certain period of time. The test for inhibiting diatom adhesion was conducted by immersing the coatings in the algae solution for a period of time, taking it out, counting the number of diatoms on the surface of the coatings and calculating the inhibiting adhesion ratio. *E. coli*, *Staphylococcus aureus* (*S. aureus*), and *Pseudoalteromonas marina* (*P. m*) were selected for the antibacterial test. The number of colonies on the surfaces of the Luria-Bertani solid culture

(1) Synthesis of Vanillin Methacrylate



(2) Synthesis of Vanillin Methacrylate random copolymer



SCHEME 1 | Scheme for the synthesis of the vanillin methacrylate (VMA) and vanillin methacrylate copolymers (VMPs) via free radical polymerization.

TABLE 1 | Formulation of the VMPs.

Samples	MMA/g	BA/g	HFBMA/g	MEA/g	VMA/g
VMP-0	57.73	31.27	9	2	0
VMP-5	54.49	29.51	9	2	5
VMP-10	51.24	27.76	9	2	10
VMP-15	47.30	26.70	9	2	15
VMP-20	44.76	24.24	9	2	20
VMP-25	41.51	22.49	9	2	25

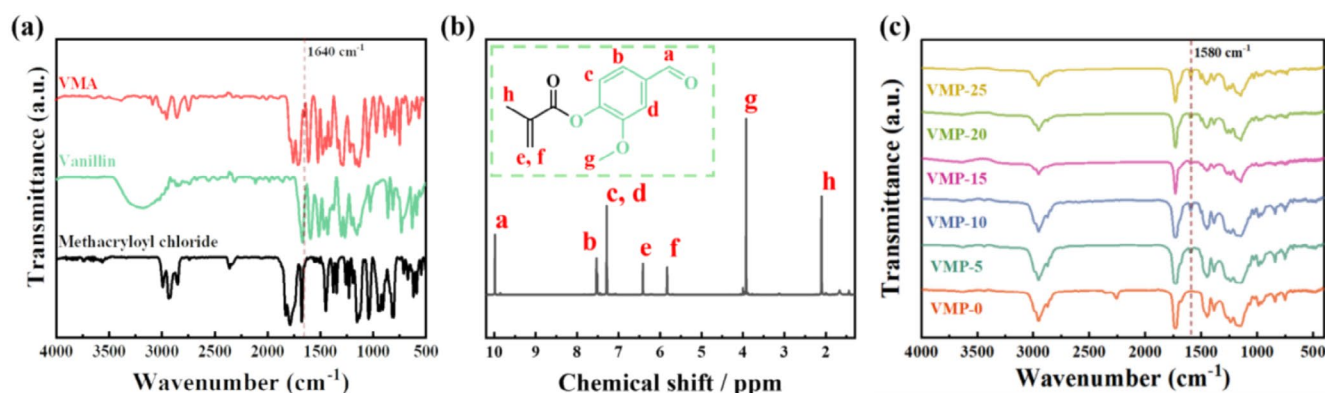


FIGURE 1 | (a) FTIR spectra of VMA, vanillin, and methacryloyl chloride, (b) ^1H NMR spectra of VMA, and (c) FTIR spectra of VMPs.

medium was counted and the antibacterial ratio was calculated. The selective adhesion test of mussels was conducted by placing multiple mussels on randomly ordered coated surfaces, observing the movement of the mussels over a period of time, and counting the number of byssus adhesive plaques on the surfaces. The marine field test was carried out at a wharf on Dachangshan Island, Dalian. Different from the laboratory environment, the VMPs resin is mixed with some fillers to prepare the coatings tested in the real sea. The formulation of the coatings used for marine field test is shown in Table S1. The calculation results were performed using Cp2k software at B3LYP/6-311++G (d, p) level. The specific characterization and test particulars are elaborated in the Supporting Information.

3 | Results and Discussion

3.1 | Characterization of VMA and VMPs

Figure 1a Shows the FTIR spectra of VMA, vanillin, and methacryloyl chloride. A new stretching vibration peak of $\text{C}=\text{C}$ was observed in VMA at 1640 cm^{-1} compared with vanillin, and the peak of $3000\text{--}3500\text{ cm}^{-1}$ belonging to the phenolic hydroxyl group in vanillin disappeared in VMA, indicating the successful introduction of methacrylate groups. Figure 1b shows the ^1H NMR spectra of VMA. The corresponding relationship between the peaks of different chemical shifts and the hydrogen atoms is as follows.

^1H NMR (500MHz, CDCl_3 , ppm): 9.97–10.01 a ($\text{CHO}-$); 7.49–7.57 b ($-\text{PhH}$), 7.26–7.32 c, d ($-\text{PhH}$); 6.38–6.44 e, 5.79–5.87 f ($-\text{C}=\text{CH}_2$), 3.90–3.95 g ($\text{CH}_3\text{O}-$), 2.08–2.13 h ($-\text{CH}_3$).

The ^1H NMR results indicate that the synthesis of VMA was successful. VMPs were characterized by FTIR, and the results are shown in Figure 1c. It can be seen that 1580cm^{-1} represents the stretching vibration peak of benzene ring and the intensity increases with the increase of VMA content, which indicates that VMA successfully participates in free radical polymerization.

3.2 | Hydrolysis Property

VMPs resin was coated on glass slides and immersed in artificial seawater for a period of time, and then taken out for characterization. The weight loss curve can directly reflect the hydrolysis rate of the coatings, as shown in Figure 2a. It is obvious that the weight loss rate of VMP-0 is the lowest, and other coatings show rapid hydrolysis weight loss in the first 7 days, which may be caused by partial residual solvent DMF and oligomer hydrolysis of a small amount of VMA monomer. After 7 days, the hydrolysis rate of the coatings tends to be stable, and the hydrolysis rate is faster with the increase of VMA content. Figure 2b

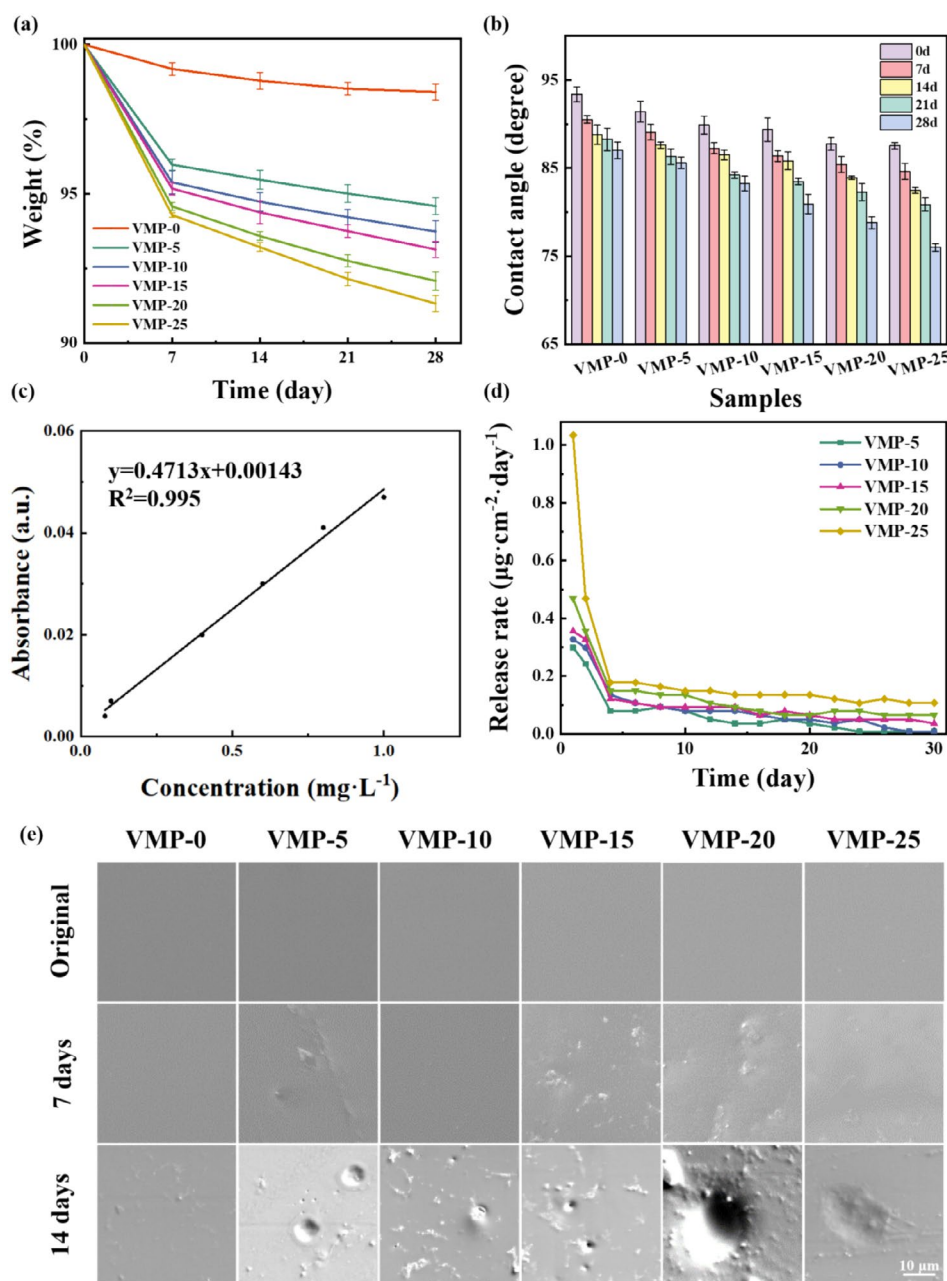


FIGURE 2 | Analysis of hydrolytic property of VMPs. (a) Weight loss curve of VMPs coatings. (b) The contact angle of coatings before and after immersion in artificial seawater. (c) The standard curve of the UV-Vis absorbance of vanillin. (d) Vanillin release rate from VMP coatings in artificial seawater. (e) SEM images of VMPs coatings before immersion and after immersion in artificial seawater.

is the change of contact angle of the coatings before and after immersing in artificial seawater. Before immersing, the contact angle of the coatings decreases with the increase of the VMA content, because the aldehyde group of VMA is hydrophilic. The better the hydrophilicity of the coatings, the faster the hydrolysis rate, ultimately shortening the service life, so the VMA content should not be too high in practical application and here we choose 25% weight fraction as the highest VMA content. In addition, the structure of vanillin makes it have a certain absorption capacity of specific wavelengths of light, which can be quantitatively detected by ultraviolet and visible spectrophotometers. In this article, the maximum UV absorption wavelength of vanillin was tested, which was 310 nm (Figure S1), and the standard curve of vanillin concentration-absorbance was fitted under this wavelength. As shown in Figure 2c, the fitted standard curve is $y = 0.4713x + 0.00143$, and the linear correlation coefficient $R^2 = 0.995$. According to the standard curve, the release rate curve of vanillin of VMPs in artificial seawater is calculated (Figure 2d). VMP-0 does not contain VMA monomer, so its release rate is 0. It can be seen that in the first 5 days, all VMP samples can release a large amount of vanillin because the rapid hydrolysis and release of oligomers of VMA monomer, which corresponds to its weight loss curve. After 5 days, VMP coatings can reach a relatively stable release rate. With the increase of immersing time, samples with high VMA content can continue to maintain stability, but samples with low VMA content become more slowly, and finally lower than the detection limit of the instrument.

SEM was used to observe the surface morphology of each coating (Figure 2e). The variance of the coating surface before and

after immersion can reflect the disparity of the hydrolysis rate. It can be seen that all coatings surfaces are smooth and flat before immersion, indicating that VMPs series resins exhibit good film-forming properties. After 7 days of immersion, the surface hydrolysis caused partial roughness and folds on the surface, while after 14 days, the surface of VMP-0 remains flat because it cannot be hydrolyzed. However, small pits appeared on the surface of VMP-5, VMP-10, and VMP-15, which was the change of coatings surface caused by hydrolysis. VMP-20 and VMP-25 showed obvious pits due to the overfast hydrolysis rate and uneven hydrolysis. This reflects that the high content of VMA is easy to self-polymerize, resulting in rapid local hydrolysis of the coatings.

3.3 | AF Properties

Benthic algae *Halophora* sp. and planktonic algae *Nitzschia closterium* were selected as typical algae to test the anti-algal adhesion performance. The blank was a glass slide. Due to the rapid growth rate and strong adhesive force of the *Halophora* sp. and the slow growth and small adhesion number of the *N. closterium*, different moments were selected for observation. The microscopy images of VMP coatings (Figure 3a) showed that the content of VMA significantly improved the adhesion inhibition effect of the VMP coatings. For *Halophora* sp., the inhibition ratio of VMP-0 was 44.4% after 12 h and only 17.5% after 72 h, while the inhibition ratio of VMP-25 was 91.7% after 12 h and remained above 80% even after 72 h (Figure 3b). Except for VMP-0, the inhibition ratio of other coatings after 24 h was slightly higher than that after 12 h. This was because

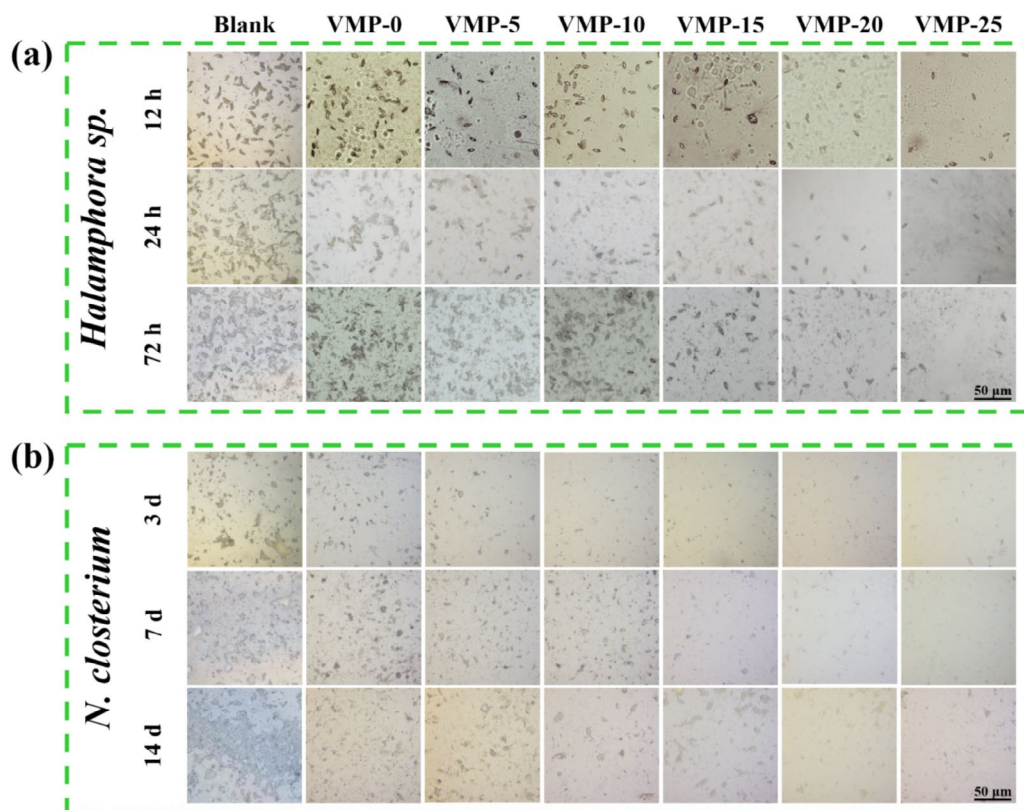


FIGURE 3 | (a) Representative microscopy images of VMP coatings immersed in *Halophora* sp. and *N. closterium* solutions. (b) *Halophora* sp. adhesion resistance ratios of VMP coatings. (c) *N. closterium* adhesion resistance ratios of VMP coatings.

the adhesion of the blank samples was faster than that of the samples (except VMP-0) and vanillin released faster at 24 h than at 12 h, resulting in an increase in the inhibition ratio of algal adhesion. Later, the growth of algae approached saturation, and the inhibition ratio began to decline. However, the inhibition ratio of *N. closterium* of all samples decreased with the increase of time due to its slow growth and long observation interval (Figure 3c). The inhibition ratio of VMP-0 was 59.2% after 3 days and decreased to 45.2% after 14 days, while VMP-25 could reach 93.6% after 3 days and maintained at 89.1% after 14 days. Through the testing of the growth curve of the *N. closterium* (Figure S2), vanillin released by the hydrolysis of VMP coatings have no significant effect on the growth of *N. closterium*, which proves the environmental friendliness of VMP coatings. It also shows that VMPs embodies the ability of anti-algae by inhibiting the adhesion of algae, rather than killing algae. The sustained release of vanillin creates a dynamically unstable surface on the VMPs coatings to prevent adhesion.

E. coli (a Gram-negative bacterium), *S. aureus* (a Gram-positive bacterium), and *P. m* (a kind of marine bacteria) were selected to conduct an antibacterial test. The result is shown in Figure 4a and b. The blank was a glass slide. Obviously, the increase in VMA content greatly improved the antibacterial ratio of the samples. VMA-0 showed worst resistance to all three types of bacteria, which was below 30%. Among them, *P. m* is more sensitive to changes in VMA content, increasing from 39.8% (VMP-5) to 92.2%, while *S. aureus* is from 48.1% (VMP-5) to 76.7% (VMP-25). VMPs are the most effective against marine

bacteria, indicating that it is more likely to realize its AF potential in the actual marine environment.

Mussels are a common large fouling organism that secretes byssus that attach to the surfaces of various underwater installations [40, 41]. The selective adhesion of mussels was tested under laboratory conditions. The result is shown in Figure 4c and d. *Mytilus edulis* was taken as the test object and placed on a glass slide coated with different coatings, and the byssus secretion was observed after a period of time. The mussels stick out their feet to find the right place to secrete byssus and adhesive plaques for adhesion (the white wireframe in Figure 4c). It is shown that in all three parallel experiments, the mussels showed a tendency to move towards the blank (glass slides) and the number of byssus plaques also reflects this trend. With the increase in VMA content, the number of byssus plaques on the coated surface decreased. This is because the coatings hydrolyze to form an unstable surface that inhibits mussel adhesion.

To research the AF performance of VMPs in real marine environment, the resin and filler were mixed evenly to prepare the coatings and carried out marine field tests in the waters of Dalian, China. As shown in Figure 5, after a four-month testing period, the blank sample surface is found to be completely covered with fouling organisms, while the VMP-0, VMP-5, and VMP-10 surface shows only a small amount of fouling. In contrast, the VMP-15, VMP-20, and VMP-25 surface exhibits minimal signs of fouling. After a period of 9 months, the blank and VMP-0 sample surfaces had been entirely covered with fouling

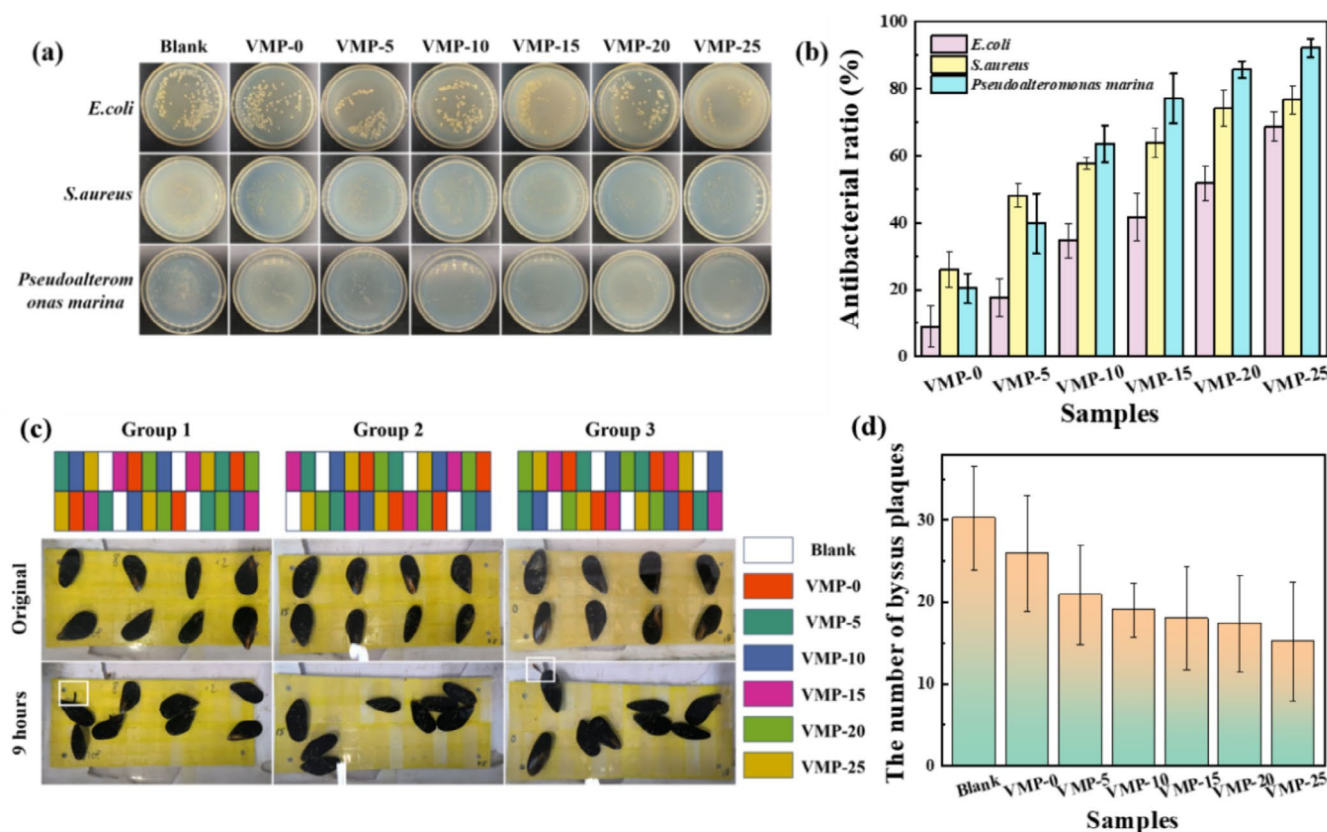


FIGURE 4 | (a) Images of antibacterial test of VMPs coatings for *E. coli*, *S. aureus*, and *P. m*. (b) Inhibition of bacterial attachment ratio of VMPs coatings. (c) Original and 9 h later mussel location. (d) The average number of byssal plaques adhered to different VMP coatings.

organisms as they were incapable of self-polishing. Nevertheless, the attachment of fouling organisms to the VMP-5, VMP-10, VMP-15, VMP-20, and VMP-25 samples demonstrated a decreasing trend, indicating that with the increase of VMA content, the AF performance of VMP coatings is better.

3.4 | Calculation of Hydrolysis Mechanism

(Methyl) acrylate resin is a prevalent self-polishing AF polymer. Its self-polishing and AF functions are mainly realized through the hydrolysis of side groups in the weak-alkaline seawater environment [12]. The hydrolysis rate of this resin is a crucial factor that significantly determines the self-polishing rate of coatings. In the weak-alkaline environment of seawater,

there are two primary hydrolysis mechanisms for ester groups, $B_{AC}2$ (OH^- attack at the carbonyl center) and S_N2 (OH^- attack at the alkyl group of the ester) mechanisms [35, 36]. The $B_{AC}2$ (base-catalyzed, acyl-oxygen cleavage, bimolecular) mechanism divides the hydrolysis process into two steps, the first step generates the tetrahedral structure of the intermediate, the second step from the tetrahedral structure decomposition to get the product. The S_N2 (base-catalyzed, alkyl-oxygen cleavage, bimolecular, also known as $B_{Al}2$) mechanism is essentially an S_N2 substitution with a carboxylate leaving group. Cp2k was shown to simulate the hydrolysis process [42]. In this study, we used Cp2k to calculate and analyze the hydrolysis mechanism of VMA and TISMA. The molecular structure is drawn in Avogadro [43] and converted to the input file of Cp2k via Multiwfn [44]. Calculations used density functional theory

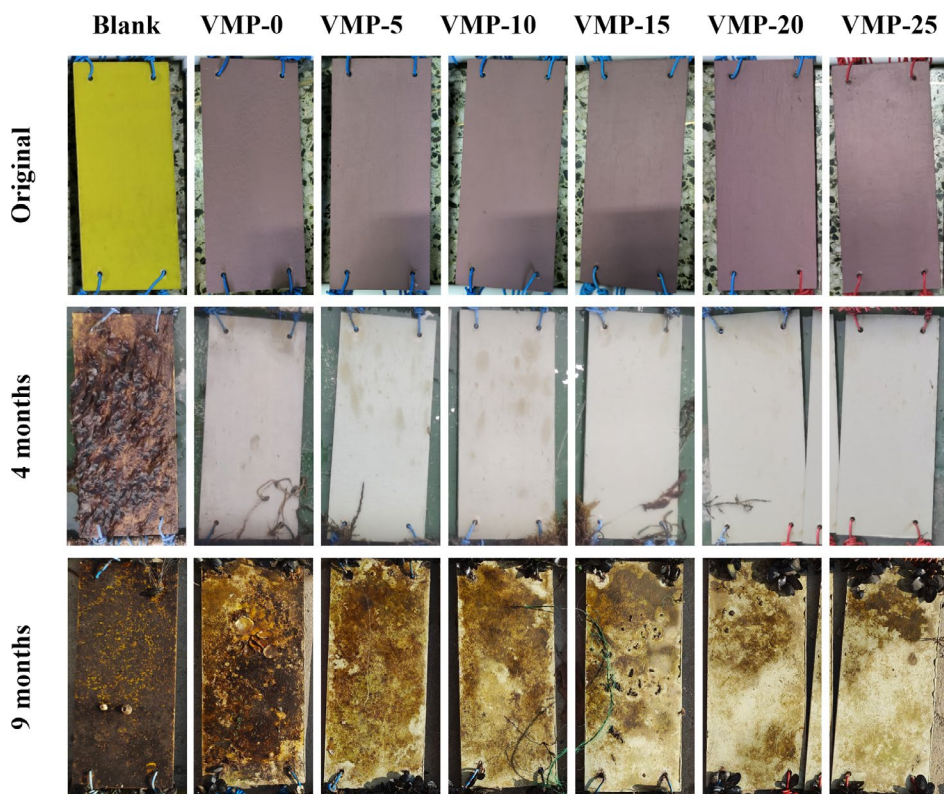


FIGURE 5 | Marine field tests images of VMPs coatings.

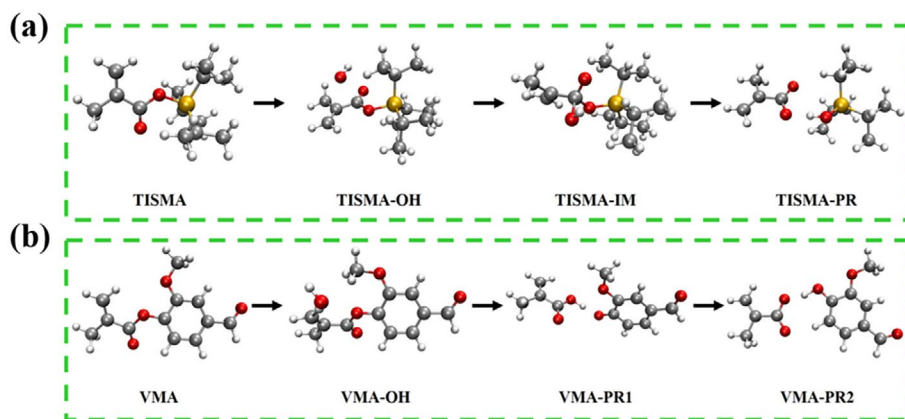


FIGURE 6 | The hydrolysis process of TISMA (a) and VMA (b).

methods at B3LYP/6-311++G (d, p) level. The result is shown in Figure 6 and the energy of the structure is shown in Table S2. Computational details and the atomic coordinates of the optimized structure are provided in supporting information. The hydrolysis process of TISMA is monomer (TISMA), monomer with OH (TISMA-OH), intermediate (TISMA-IM), and product (TISMA-PR), while the hydrolysis process of VMA is monomer (VMA), monomer with OH (VMA-OH), product 1 (VMA-PR1), and product 2 (VMA-PR2). We found that for TISMA, it can form a stable tetrahedral structure (Figure 6a), which indicates that the hydrolysis reaction is divided into two steps, and the intermediate is sandwiched between the two reaction energy barriers, which is the local energy minimum point. It means that TISMA is more prone to hydrolysis by the $B_{AC}2$ mechanism. Although VMA is different, we tried it many times, but it does not form a stable tetrahedral structure (Figure 6b). We suspect that the formation of new chemical bonds between carbonyl carbon atoms and OH and the breaking of old chemical bonds between carbonyl carbon atoms and OPh group occurred simultaneously, which is similar to the process of S_N2 reaction. Then the H from the carboxyl group is transferred to the phenol hydroxyl group, and finally forms vanillin. This means that VMA has a good leaving group (vanillin) and TISMA has a worse leaving group (triisopropylsilanol) [45]. This may be one of the reasons for the poor AF effect of TISMA-based acrylates in static condition. On the other hand, monomers containing phenolic ester groups such as VMA can be directly hydrolyzed and release compounds containing phenolic hydroxyl groups, which are found to be functional groups with AF activity [17].

4 | Conclusions

To sum up, we have successfully prepared a new and eco-friendly self-polishing AF resin that can actively release vanillin via hydrolysis of phenol ester groups in seawater. With the release of vanillin, a dynamically unstable surface is generated, endowing the VMP coatings with superior AF properties. The increase in VMA content markedly enhanced the hydrolysis rate of the coating, the release of vanillin, as well as its antibacterial properties and efficacy in inhibiting the adhesion of diatoms and mussels. The VMA hydrolysis mechanism calculated by Cp2k software proves the feasibility of self-polishing of phenol ester hydrolysis and the hydrolysis mechanism is different from TISMA, which may mean shorter onset time and better static anti-fouling performance. The same environmentally friendly, easier hydrolysis process, and better AF properties make VMA a potential alternative to TISMA. This study demonstrates the promising development potential of phenol ester group structures in marine AF and offers robust support for subsequent monomer screening and hydrolysis rate prediction, presenting a novel concept for the advancement of environmentally friendly marine AF coatings.

Acknowledgments

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Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.